

Trastuzumab deruxtecan (Enhertu) in Breast Cancer

Drug Summary

Trastuzumab deruxtecan (also known as DS-8201a, fam-trastuzumab deruxtecan-nxki, or T-DXd) is a HER2-directed antibody–drug conjugate (ADC) approved for advanced human epidermal growth factor receptor 2 (HER2)–positive breast cancer that is unresectable or metastatic after prior anti-HER2 therapy ¹. It is also approved for “HER2-low” breast cancer (IHC 1+ or 2+/ISH–) after prior chemotherapy, including HR-positive or HER2-ultralow (IHC 0/faint) tumors that have progressed on endocrine therapy ² ³. Key trials demonstrated superior efficacy over standard therapy: in DESTINY-Breast03 (HER2+ MBC, post-trastuzumab/taxane) T-DXd dramatically improved progression-free survival (12-month PFS ~76% vs 34% with T-DM1; HR 0.28) and overall response rate ⁴. In DESTINY-Breast04 (HER2-low MBC) T-DXd significantly prolonged PFS and overall survival versus chemotherapy (median PFS 9.9 vs 5.1 months; HR≈0.50 for progression or death) ⁵. T-DXd’s mechanism – a humanized anti-HER2 IgG1 conjugated via cleavable linker to a topoisomerase-I inhibitor (DXd) – allows targeted delivery of DNA-damaging payload into HER2-expressing tumor cells ⁶. The trastuzumab moiety also engages immune effector mechanisms (e.g. Fcγ receptor–mediated ADCC/ADCP) ⁶. Clinically, T-DXd is now a standard later-line therapy for HER2+ MBC and for endocrine-refractory or chemotherapy-refractory HER2-low metastatic breast cancer ¹ ². Notably, T-DXd has activity in both hormone-receptor–positive and triple-negative subtypes, provided they meet the HER2 expression criteria ² ⁵.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL4297844 ⁷.
- **DrugBank ID:** DB14962 ⁸.
- **CAS #:** 1826843-81-5 ⁷.
- **UNII:** 7I1EK1PHC4 (fam-trastuzumab deruxtecan-nxki) ⁷.
- **Synonyms:** fam-trastuzumab deruxtecan-nxki; trastuzumab deruxtecan (USAN/INN) ⁹; DS-8201, DS-8201A, DS8201, DS8201a (development codes) ¹⁰; T-DXd or TDXd ⁹; brand name **Enhertu** ¹¹. All identifiers and synonyms are confirmed by DrugBank or ChEMBL databases ⁷ ⁹.

Mechanism of Action (Breast Cancer Context)

Trastuzumab deruxtecan is an ADC combining a humanized anti-HER2 (ERBB2) IgG1 antibody with deruxtecan (DXd), a potent topoisomerase I inhibitor ⁶. Upon binding to HER2 on tumor cells, the ADC is internalized; lysosomal enzymes cleave the peptide linker and release DXd inside the cell ⁶. DXd then induces targeted DNA damage (single-strand nicks that convert to double-strand breaks), triggering apoptosis ⁶. The cleavable linker and membrane-permeable payload permit a “bystander effect,” where neighboring HER2-low cancer cells can be killed by diffusing DXd, although in vitro studies show this effect is modest and requires substantial HER2+ cell presence ¹² ¹³. In breast cancer cells, this mechanism synergizes with HER2 signaling dependence: HER2-overexpressing cells avidly internalize T-DXd and are

highly sensitive ⁶ . The trastuzumab antibody also retains Fc-mediated effects: it can engage high-affinity Fcγ receptors (e.g. FCGR1A on macrophages) to induce antibody-dependent cellular phagocytosis (ADCP) ¹⁴ ¹⁵ and may elicit ADCC. T-DXd's cytotoxic payload also induces immunogenic cell death (ICD): studies show T-DXd-treated tumor cells activate the STING (IFN-I) and TLR4 pathways in antigen-presenting cells, enhancing type-I interferon responses and CD8⁺ T-cell activation ¹⁶ ¹⁷ . However, T-DXd can upregulate the "don't eat me" signal CD47 on tumor cells, limiting phagocytosis ¹⁸ . In sum, T-DXd acts by targeted HER2 binding, intracellular release of a topo-I poison, induction of DNA damage/apoptosis, and engagement of immune pathways (IFN, FcγR). HER2 expression is required for activity – loss of HER2 causes resistance ¹⁹ – and HDRD/BRCA status may modulate response via DNA-repair capacity (see Sensitivity/Resistance below).

Primary Targets (Human)

- **ERBB2 (HER2):** The antibody component binds the extracellular domain of ERBB2 ⁶ . HER2 overexpression drives oncogenic signaling in ~15–30% of breast cancers ²⁰ ; T-DXd targets HER2 as its primary tumor antigen.
 - **TOP1 (DNA Topoisomerase I):** The DXd payload is a potent inhibitor/poison of TOP1 ⁶ , trapping the cleavable complex and causing DNA strand breaks.
 - **FCGR1A (High-affinity IgG Fcγ Receptor I):** Trastuzumab's Fc region can bind FCGR1A on immune cells (macrophages, etc.) to mediate ADCP ²¹ ¹⁵ . This is a validated interaction for T-DXd (FDA labeling notes high-affinity FcR binding ²¹).
- These targets are experimentally supported by DrugBank data and mechanistic studies ⁶ ²¹ .

Pathways (High-Level Overview)

Key cancer pathways perturbed by trastuzumab deruxtecan include:

- **ERBB2/HER2 signaling** (e.g. Reactome *Signaling by ERBB2 in Cancer*, R-HSA-1227990). HER2 drives proliferation via RAS/MAPK and PI3K/AKT/mTOR pathways. T-DXd's antibody blocks HER2 and downmodulates this pathway. Loss of ERBB2 signaling (HER2 downregulation or mutation) leads to treatment resistance ¹⁹ , while HER2 amplification sensitizes to the drug.
- **PI3K/AKT/mTOR signaling** (KEGG hsa04151). A major downstream effector of HER2. Inhibition of HER2 by T-DXd indirectly suppresses PI3K/AKT; conversely, upregulation of PI3K/AKT can promote survival signaling. This pathway is associated with HER2-driven proliferation.
- **Cell cycle / G2-M checkpoint** (MSigDB *HALLMARK_G2M_CHECKPOINT*, Reactome *Mitotic G2/M phases*). Topo-I inhibitors induce DNA damage that activates the G2/M checkpoint. Resistance studies found this pathway upregulated in T-DXd-resistant cells ²² .
- **DNA Repair / Damage Response** (MSigDB *HALLMARK_DNA_REPAIR*, KEGG *Homologous Recombination* hsa03440). DXd causes DNA breaks; activation of DNA repair pathways (ATR/ATM/BRCA1/2, homologous recombination, NHEJ) influences sensitivity. Upregulation of DNA repair genes (e.g. RAD21, MCL1 amplification) is seen in resistance ²² , while inhibiting repair (e.g. ATR inhibition) enhances T-DXd efficacy ²³ .
- **Apoptosis** (MSigDB *HALLMARK_APOPTOSIS*, Reactome *Programmed Cell Death*). DXd induces apoptosis via DNA damage. Upregulation of anti-apoptotic factors (e.g. MCL1) confers resistance ²⁴ .
- **FcγR-mediated phagocytosis** (Reactome *FCGR dependent phagocytosis*, R-HSA-2029480). The Fc portion of trastuzumab engages Fcγ receptors on myeloid cells, promoting ADC uptake and tumor cell killing ⁶ ¹⁵ . This immune effector pathway contributes to antitumor activity.

- **Innate immune sensing (STING/IFN-I and TLR4)** (Reactome *STING-mediated induction of host immune responses*, R-HSA-1834941; MSigDB *HALLMARK_INTERFERON_ALPHA_RESPONSE*). T-DXd-treated tumors activate STING and type-I interferon signaling, enhancing dendritic cell and CD8⁺ T-cell responses ¹⁶. Genes in the TLR4–STING–IFN axis are upregulated after T-DXd-induced ICD ¹⁶.
- **Inflammatory/immune signaling (IL6–JAK–STAT3, IFN γ)** (MSigDB *HALLMARK_IL6_JAK_STAT3_SIGNALING*, *HALLMARK_INTERFERON_GAMMA_RESPONSE*). Enrichment of these pathways has been observed in patients with active immune microenvironments ²⁵. A pre-existing interferon signature (IFN α/γ , IL6/JAK-STAT) was higher in T-DXd responders ²⁵.
- **TNF/NF- κ B signaling** (MSigDB *HALLMARK_TNFA_SIGNALING_VIA_NFKB*). Chronic inflammatory signaling may modulate ADC response; in preclinical models, TNF blockade enhanced T-DXd by amplifying innate immunity (via M1 macrophages) ²⁶.
- **Cell migration/EMT pathways**. While not a direct target, ADC-induced changes can alter epithelial–mesenchymal transition markers; some resistance phenotypes involve EMT, but evidence is limited.

These pathways span the antibody target (HER2/ERBB2) and its downstream signals (PI3K/AKT, cell cycle), the payload's action (DNA repair, apoptosis), and immune components (Fc γ R phagocytosis, STING–IFN, cytokine signaling). Pathway annotations and gene-set enrichments (from MSigDB and Reactome) highlight how upregulation or suppression of these processes can affect T-DXd sensitivity or resistance.

Upregulated Pathways

Studies of tumor samples and cell lines exposed to T-DXd reveal certain pathways are transcriptionally **upregulated**:

- **DNA Damage Response / DNA Repair** (MSigDB *HALLMARK_DNA_REPAIR*): In T-DXd-resistant HER2+ breast cancer models, genes in DNA repair pathways (e.g. homologous recombination, NHEJ) were significantly upregulated ²² ²⁷. Tumors post-anti-HER2 therapy showed amplifications of DNA repair genes (TOP2A, RAD21, RAD52) ²⁸, suggesting enhanced repair activity as an adaptive response.
- **G2/M Checkpoint** (MSigDB *HALLMARK_G2M_CHECKPOINT*): Genes controlling G2-to-M phase were enriched in resistant cells ²². Upregulation of this checkpoint may allow tumor cells to survive and repair DXd-induced DNA damage before mitosis.
- **Mitotic Spindle** (MSigDB *HALLMARK_MITOTIC_SPINDLE*): Also activated in T-DXd-resistant lines ²², reflecting heightened proliferation signaling under drug pressure.
- **Interferon Responses** (MSigDB *HALLMARK_INTERFERON_GAMMA_RESPONSE* and *ALPHA_RESPONSE*): Tumors with an active immune microenvironment showed enrichment of interferon- α/γ genes prior to treatment ²⁵. While not induced by T-DXd per se, this pre-existing upregulation correlates with better drug activity (see Sensitivity).
- **IL6–JAK–STAT3** (MSigDB *HALLMARK_IL6_JAK_STAT3_SIGNALING*): Similarly enriched in immune-active tumors ²⁵, reflecting inflammatory signaling that may cooperate with T-DXd's immunogenic effects.
- **Fc γ R signaling**: In immune cells co-cultured with T-DXd-treated tumors, Fc γ R pathways were upregulated as macrophages engaged opsonized cells ¹⁵.
- **TLR4 and STING Pathways**: RNA-seq of macrophages exposed to T-DXd-treated tumor cells showed marked upregulation of TLR4 signaling genes and interferon-stimulated genes ¹⁷, confirming induction of innate immune sensing after ADC therapy.

Downregulated Pathways

Pathways found **downregulated** in T-DXd-treated or resistant contexts include:

- **ERBB2/HER2 Signaling (Reactome *Signaling by ERBB2 in Cancer*)**: The most notable is downregulation of HER2 itself. In nearly half of patients who progressed on T-DXd, HER2 expression was greatly reduced or lost ¹⁹. Loss of ERBB2 gene amplification or expression abrogates T-DXd binding and thus is associated with resistance.
- **Apoptosis (MSigDB *HALLMARK_APOPTOSIS*)**: Resistant cells often upregulate anti-apoptotic factors (MCL1, BCL-2 family) and downregulate pro-apoptotic signals ²⁴. This effective suppression of apoptosis machinery accompanies treatment failure.
- **Immune Activation Signatures**: Some resistant tumors show decreased immune genes. For example, lack of interferon/IL6 signaling (i.e. low *IFN γ _RESPONSE*) would reflect an immunosuppressive microenvironment, possibly reducing bystander immune effects.
- **General Metabolism/Growth**: Although less documented, chronic drug pressure can downregulate proliferative pathways in surviving cells (analogous to EMT changes), further diminishing drug impact.

Sensitivity Pathways (Upregulation)

Pathway activities whose **upregulation** increases T-DXd sensitivity (often by enhancing drug entry or immune effect):

- **ERBB2/HER2 Signaling**: Upregulation means higher HER2 expression/amplification, providing more ADC binding sites. HER2-overexpressing (ERBB2-amplified) tumors are highly sensitive to T-DXd ⁶. Clinically, HER2 gene amplification (and the intrinsic “HER2-enriched” subtype) predicts exceptional response to HER2-directed therapy. In short, strong ERBB2 signaling (upregulated pathway) is required for T-DXd efficacy ¹⁹.
- **Fc γ R-mediated Phagocytosis**: Increased expression of activating Fc γ receptors (e.g. FCGR1A on macrophages) or enhanced ADCC pathway activity would improve clearance of ADC-opsonized cells. While direct clinical evidence is limited, the mechanism implies that tumors with high macrophage infiltration and Fc γ R activity may be more susceptible to T-DXd.
- **Interferon/Innate Immune Pathways**: Upregulation of the STING/IFN-I and TLR4 pathways in the tumor microenvironment can amplify T-DXd’s antitumor immune effects. T-DXd induces immunogenic cell death that activates STING and TLR4 in antigen-presenting cells ¹⁶; thus, a pre-existing upregulation of these innate immune pathways (or the ability to upregulate them) should sensitize tumors by boosting CD8⁺ T-cell responses.
- **Interferon Gamma Response (HALLMARK_IFN γ)**: Upregulation of IFN γ -related genes (e.g. MHC, chemokines) in tumors is associated with “hot” immune microenvironment. In DESTINY-Breast04, patients with higher baseline interferon signatures had better outcomes ²⁵, suggesting that an IFN γ -active state sensitizes tumors to the immune component of T-DXd therapy.
- **Cell-Cycle (G1/S transition)**: Rapidly proliferating cells (with high E2F/cell-cycle activity) may be more vulnerable to topoisomerase inhibitors during S-phase. Although specific markers are not established for T-DXd, in general upregulation of S-phase entry could increase payload incorporation into DNA.

Sensitivity Pathways (Downregulation)

Pathway activities whose **downregulation** increases T-DXd sensitivity:

- **DNA Repair / ATR Pathway**: Reduced DNA repair capacity is expected to sensitize cells to DXd-induced

damage. Loss-of-function in HR genes (e.g. BRCA1/2, RAD51) or pharmacologic inhibition of ATR/CHK1 massively increases T-DXd potency. A recent preclinical study showed that inhibiting the ATR-mediated DNA damage response synergized with T-DXd in resistant HER2+ models ²³. Thus, downregulation of ATR/ATM/ DNA-repair signaling (and HRD status) is predicted to enhance efficacy.

- **PI3K/AKT Signaling:** Downregulation of PI3K/AKT survival signaling could sensitize tumors by promoting apoptosis. For example, tumors with PTEN loss (PI3K overactive) may be more resistant, whereas upregulation of phosphatases or PI3K inhibitors (downstream) might improve killing. (Note: This is hypothesized; specific evidence with T-DXd is limited.)

- **Cell Cycle Checkpoints:** Reduction of the G2/M checkpoint (e.g. via WEE1 inhibition) forces cells with T-DXd-induced damage into lethal mitosis. Indeed, combining T-DXd with a WEE1 inhibitor showed enhanced cancer cell death in vitro ²⁹ ³⁰. Thus, downregulation of cell-cycle arrest pathways (ATR/CHK1/WEE1) increases sensitivity.

- **Anti-Apoptotic Signals:** Downregulating anti-apoptotic pathways (e.g. BCL2, MCL1) unleashes DXd-induced apoptosis. Though not directly studied with T-DXd, lowering MCL1 levels is a known strategy to sensitize to DNA-damaging agents.

- **Immune Checkpoints (CD47):** Although an “immune checkpoint” rather than a classic pathway, downregulation or blockade of tumor CD47 (the “don’t eat me” signal) markedly enhances T-DXd activity via macrophage phagocytosis ¹⁸. In other words, suppressing CD47 signaling increases drug sensitivity.

Resistance Pathways (Upregulation)

Pathway activities whose **upregulation** causes T-DXd resistance by counteracting its mechanisms:

- **DNA Repair / ATR-CHK1 Pathways:** Upregulation of DDR pathways is a key resistance mechanism. Resistant tumors amplify genes like RAD21, RAD52, and especially ATR/CHK1 signaling components ²² ³¹. This enhanced repair capability allows cancer cells to fix DXd-induced DNA breaks. Correspondingly, clinical-resistant samples showed amplified DNA repair genes after therapy ²⁸.

- **Cell-Cycle Checkpoints (G2/M):** Overactivation of G2/M checkpoint and mitotic spindle genes was observed in T-DXd-resistant cell lines ²². By arresting the cell cycle, these tumors avoid lethal mitosis and repair DNA damage.

- **Anti-Apoptotic / Survival Signaling:** Upregulation of anti-apoptotic factors (e.g. MCL1, BCL2) confers resistance. In one patient series, MCL1 was amplified in resistant tumors ²⁸. Additionally, heightened PI3K/ AKT or MAPK signaling (downstream of receptor tyrosine kinases) can promote survival despite DNA damage.

- **FcyR Signaling Desensitization:** Tumors may adapt by upregulating FcyR inhibitory pathways or ligands. Notably, T-DXd-killing induces tumor cell expression of CD47 ¹⁸, dampening macrophage phagocytosis. Thus, upregulated CD47–SIRPα signaling acts as an evasion mechanism (resistance) to the ADC’s immune component.

- **Drug Target Mutation/Alteration:** Although not a “pathway” per se, activating mutations in ERBB2 that disrupt trastuzumab binding (e.g. V597M in HER2) have been reported to arise under ADC therapy ³². This effectively “upregulates” constitutive ERBB2 signaling while preventing ADC binding, causing resistance.

- **Stromal/ECM Pathways:** Increased expression of extracellular matrix components or efflux transporters could physically impede ADC penetration. (Direct evidence is sparse, but increased ECM signatures often correlate with drug resistance in solid tumors.)

Resistance Pathways (Downregulation)

Pathway activities whose **downregulation** leads to T-DXd resistance:

- **ERBB2/HER2 Signaling:** The single most critical case is **loss of HER2 expression**. Tumors with reduced ERBB2 copy number or IHC score can no longer internalize the ADC, leading to abrupt resistance ¹⁹. Indeed, ~50% of biopsies at progression on T-DXd showed marked HER2 downregulation or loss ¹⁹.
- **FcyR and Innate Immune Pathways:** Downregulation of macrophage or NK cell FcyR signaling would blunt the immune-mediated component of T-DXd action. Experimentally, lack of TLR4 or STING in macrophages abolished T-DXd-induced innate activation ³³. Thus, reduced expression of TLR4–STING–IFN pathways (or host immune defects) could resist the immunogenic effects.
- **DNA Damage Sensing:** Paradoxically, reduction of pro-death pathways (e.g. TP53/p53 downregulation) can confer chemoresistance. If p53-mediated apoptosis is impaired, even high DNA damage may not kill cancer cells effectively.
- **Other Receptors:** Downregulation of molecules involved in ADC uptake (e.g. lysosomal transporters, cell adhesion receptors) could theoretically cause resistance, but specific examples in T-DXd are not documented.

Subtype-Specific Evidence

- **HER2-positive (IHC3+) Breast Cancer:** T-DXd is highly active. The DESTINY-Breast03 trial in HER2+ MBC (second-line) showed clear PFS and ORR benefit over T-DM1 ⁴. Intrinsic HER2-enriched tumors (often luminal B or HER2 subtype by PAM50) are particularly sensitive due to abundant HER2 ²⁰. Resistance in this subgroup often involves acquired HER2 loss ¹⁹.
- **HR-positive, HER2-negative (Luminal A/B):** In this broad category, only *HER2-low* tumors benefit from T-DXd. DESTINY-Breast04 established efficacy in HR+/HER2-low metastatic breast cancer ⁵. U.S. labeling further specifies HR+ HER2-low or ultralow tumors after endocrine failure ³⁴. In contrast, true HR+ HER2- (IHC0) without HER2 expression gain no direct benefit.
- **Triple-negative (TNBC) / Basal-like:** Approximately 30–40% of DESTINY-Breast04 patients had TNBC (HER2-low). These TNBC cases derived similar PFS/OS benefit as HR+ (T-DXd vs chemo) ⁵. TNBC is often basal-like; such tumors with low HER2 still respond. Ongoing trials are exploring T-DXd ± immunotherapy in TNBC. Notably, PD-L1 expression has not precluded benefit, though one retrospective report noted PD-L1–negative/BRCA-mutated patients had worse outcomes **[63†]** (data still emerging).
- **gBRCA-mutated / HRD-positive:** The impact of germline BRCA1/2 on T-DXd efficacy is unclear. One real-world report suggested BRCA-mutant patients had *no* enhanced benefit (possibly worse) **[63†]**, implying that HER2 expression (not HRD) dominates sensitivity. However, preclinical rationale suggests HRD tumors might be more vulnerable to DNA-damaging payloads ²³. Clinical data are limited, so no definitive guidance.
- **Basal-like Subtype:** As basal-like largely overlaps TNBC, the above TNBC findings apply. Some basal-like tumors express low HER2 and would benefit similarly.
- **HER2-enriched Subtype:** Tumors classified as HER2-enriched by gene expression (often HER2-amplified) show robust T-DXd responses akin to classic HER2+ disease. These often have high HER2/ERBB2 signaling and thus high sensitivity.

Contraindications and Safety

According to regulatory labeling, **no formal contraindications** (aside from known hypersensitivity to components) are listed ³⁵. However, major safety concerns include:

- **Interstitial Lung Disease (ILD)/Pneumonitis:** Boxed warning. ILD (including fatal cases) occurred in ~10–12% of patients (all grades) ³⁶. Cases can be severe; T-DXd should be discontinued permanently for grade ≥ 2 ILD ³⁷.

- **Embryo-Fetal Toxicity:** Boxed warning. T-DXd causes embryo-fetal harm; contraception is required during and after treatment ³⁸.

- **Myelosuppression:** Neutropenia is common ($\approx 50\%$ any grade) ³⁹; monitor blood counts. Grade 3–4 neutropenia may necessitate dose delays/reductions.

- **Cardiotoxicity:** Left ventricular dysfunction can occur ⁴⁰. LVEF should be assessed at baseline and periodically; discontinue for symptomatic CHF or significant LVEF drop.

- **Other toxicities:** Nausea/vomiting, anemia, thrombocytopenia, fatigue, alopecia, and liver enzyme elevations (AST/ALT) are frequent ⁴¹. Alopecia and gastrointestinal symptoms are seen in ~20–30% of patients ⁴¹.

- **Dermatologic:** Alopecia, dry skin, and rash may occur (similar to other topoisomerase inhibitors).

- **QT prolongation:** Not a prominent issue in trials, but caution in patients on QT-prolonging drugs.

Dose modifications or interruption are recommended for severe toxicities (ILD, neutropenia, cardiotoxicity) ⁴² ⁴³. FDA labeling and NCCN guidelines emphasize vigilant monitoring for ILD/pneumonitis, hematologic counts, and cardiac function. Safety data from EMA/Japan are concordant, with no new contraindications. Overall, T-DXd's safety profile is manageable with appropriate monitoring ⁴⁴.

Clinical Trials and Guidelines

T-DXd's approval and use are guided by pivotal clinical trials and expert recommendations. In addition to DESTINY-Breast03 ⁴ and DESTINY-Breast04 ⁵, ongoing studies continue to refine its role. For example, DESTINY-Breast05 is evaluating T-DXd in HER2+ early breast cancer. Phase I/II trials are exploring T-DXd in HER2-negative subtypes (HER2-mutant BC, TNBC with low HER2, etc.). Guidelines (NCCN, ESMO, ASCO) have incorporated T-DXd for metastatic HER2+ disease after 1–2 prior lines, and for HER2-low metastatic disease after standard endocrine or chemo. For instance, the 2024 NCCN Breast Cancer Guidelines list Enhertu for unresectable/metastatic HER2+ tumors after prior trastuzumab and taxane, and for HR+ HER2-low tumors after endocrine therapy ³. These recommendations are backed by the cited trials. Additionally, exploratory research (e.g. non-small-cell lung and gastric cancer studies) demonstrates the broader principle of HER2-targeted ADCs, but for breast cancer the evidence is strongest in the above settings.

Additional Notes

- **Bystander Effect:** Preclinical models confirm that DXd can kill neighboring HER2-low cells when adjacent to HER2-high cells, but this bystander cytotoxicity is quantitatively limited ¹². Clinical efficacy in HER2-low patients suggests some bystander killing or sufficient low-level uptake in low-HER2 cells.
- **Immune Synergy:** Combining T-DXd with immune therapies is under investigation. Notably, T-DXd upregulates tumor PD-L1 and CD47 ¹⁸; trials are testing PD-1/PD-L1 or CD47 inhibitors with T-DXd to overcome immune resistance.

- **Biomarkers:** Circulating HER2 (ctDNA) and imaging (e.g. HER2 PET) are being studied to predict response. Early data suggest that HER2 heterogeneity on PET may correlate with T-DXd benefit ¹³.
- **Pharmacokinetics:** The DXd payload is metabolized by CYP3A4 ⁴⁵, so strong CYP3A inhibitors/inducers should be used cautiously (per label).
- **Drug Interactions:** No major drug–drug interactions have been highlighted, but avoid severe CYP3A modulation due to the DXd component.

Summary of Mechanistic Evidence

Pathway (ID/Name)	Regulation	Effect	Rationale	Ref.
ERBB2/HER2 signaling (e.g. R-HSA-1227990)	Down	Resistant	Loss or downregulation of HER2 (ERBB2) prevents ADC binding/internalization, causing resistance ¹⁹ .	¹⁹
ERBB2/HER2 signaling	Up	Sensitive	HER2 overexpression (amplification) provides target; upregulation of ERBB2 signaling is required for ADC efficacy ⁶ .	⁶ ¹⁹
DNA repair (HALLMARK_DNA_REPAIR)	Up	Resistant	Upregulation of homologous recombination/NHEJ genes (e.g. RAD21, RAD52, MCL1) in resistant cells enhances repair of DXd-induced damage ²² .	²²
DNA repair (HALLMARK_DNA_REPAIR)	Down	Sensitive	Defective DNA repair (e.g. HRD from BRCA loss, ATR/CHK1 inhibition) impairs damage resolution, increasing DXd cytotoxicity ²³ .	²³
G2/M checkpoint (HALLMARK_G2M_CHECKPOINT)	Up	Resistant	Enhanced G2/M checkpoint activity allows cell-cycle arrest and DNA repair under drug, promoting survival ²² .	²²
FcγR-mediated phagocytosis (R-HSA-2029480)	Up	Sensitive	Upregulation of FcγR (e.g. FCGR1A) enhances antibody-dependent phagocytosis of ADC-opsonized tumor cells ⁶ ¹⁵ .	⁶ ¹⁵

Pathway (ID/Name)	Regulation	Effect	Rationale	Ref.
STING/IFN-I pathway (R-HSA-1834941)	Up	Sensitive	T-DXd induces immunogenic cell death activating STING and IFN- α ; upregulation of this pathway amplifies innate/adaptive immunity ¹⁶ .	¹⁶
CD47-SIRP α signaling	Up	Resistant	T-DXd-induced upregulation of tumor CD47 inhibits macrophage phagocytosis (immune escape); high CD47 confers resistance ¹⁸ .	¹⁸
IFN γ response (HALLMARK_INTERFERON_GAMMA)	Up	Sensitive	Pre-existing high IFN γ signaling (immune-hot tumors) correlates with better T-DXd response, suggesting immune activation aids efficacy ²⁵ .	²⁵
Anti-apoptosis (e.g. MCL1)	Up	Resistant	Upregulated anti-apoptotic genes (MCL1 amplification) were observed in resistant tumors, blocking DXd-induced apoptosis ²⁴ .	²⁴
PI3K/AKT pathway (HALLMARK_PI3K_AKT_MTOR)	—	Mixed*	Downstream of HER2; upregulation promotes survival (resistance), whereas pathway inhibition sensitizes to apoptosis (sensitivity) ⁶ ²³ . (Dual role.)	⁶ ²³

*"Mixed" indicates context-dependent effects. In practice, activated PI3K/AKT signaling can drive resistance by promoting survival, whereas its downregulation (or inhibition) facilitates T-DXd-induced apoptosis.

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